金逸晨电子有限公司

ROUND TFT LCD MODULE

1.28 inch 240RGB*240DOTS

MODULE NUMBER: GMT128-02

REVISION: **A**

Customer:
Approved by

From: Golden Morning Technology Co.,Ltd
Approved by

Notes:

1. Please contact Golden Morning Technology Inc. before assigning your product based on this module specification
2. The information contained herein is presented merely to indicate the characteristics and performance of our products. No responsibility is assumed by Limito Technology Inc. for any intellectual property claims or other problems that may result from application based on the module described herein.

Revised History

Part Number	Revision	Revision Content	Revised on
GMT128-02	A	New	2022-02-26

Contents

1. General Description

1.1 Description

LH128R-IG01 is a 240RGBX240 dot-matrix TFT LCD module. This module is composed of a TFT LCD Panel, driver ICs, FPC and a Backlight unit.

1.2 Features

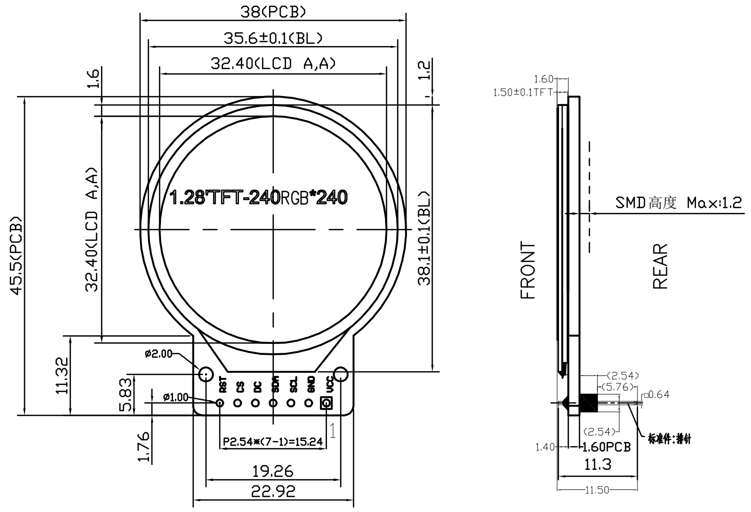
NO.	Item	Contents	Unit
1	LCD Size	1.28	inch
2	Display Mode	Normally black	-
3	Resolution	240(H)RGB x 240(V)	pixels
4	Dot pitch	0.135(H) x 0.135(V)	mm
5	Active area	Ø32.4	mm
6	Module size	38.00(H) x 45.50(V) x 11.50(D)	mm
7	Color arrangement	RGB Vtertical stripe	-
8	Interface	4 Line SPI	-
9	Drive IC	GC9A01	-
10	Luminance(cd/m2)	400 (TYP)	Cd/m2
11	Viewing Direction	All View	O'Clock
12	Backlight	2 White LED Parallel	-
13	Operating Temp.	-20℃~ + 70℃	℃
14	Storage Temp.	-30℃~+ 80℃	℃
15	Weight	TBD	g

金逸晨电子有限公司

2. Mechanical Drawing

VER.	REVISED RECORD	DATA
V1.0	第一次发行	2022-02-26

1.外形图



2.功能&特性

2-1. 点阵: 240RGBX240 Dots
2-2. 显示类型: 1.28" TFT
2-3. 视角方向: ALL
2-4. POLARIZER MODE: TRANSMISSIVE/NORMALLY BLACK
2-5. LCM Luminance:400 CD/M2(TYP)
2-6. BACK LIGHT(背光): 2 CHIP-WHITE LED
If=40mA Vf=2.9~3.1V
2-7. 工作温度: -20℃~70℃
2-8. 储存温度: -30℃~80℃
2-9. 连接方式/驱动IC:COG/GC9A01
2-10.接口方式: 4-SPI
2-11.建议机壳开窗可视区比 LCD A.A区单边大 0.3mm

3.机械规格

3-1. 模块尺寸: 38.00(H) x 45.50(V) x 11.50 Max(D)
3-2. 有效区域(A/A): ø32.4mm
3-3. 点距离: 0.21mm(L)*0.21mm(W)

PIN	DESCRIPTION
1	VCC
2	GND
3	SCL
4	SDA
5	DC
6	CS
7	RST

SHEET: 1 of 1	金逸晨电子有限公司

APPROVALS		DATE
DWN	ZJP	2022-02-26
CHK		
APP		

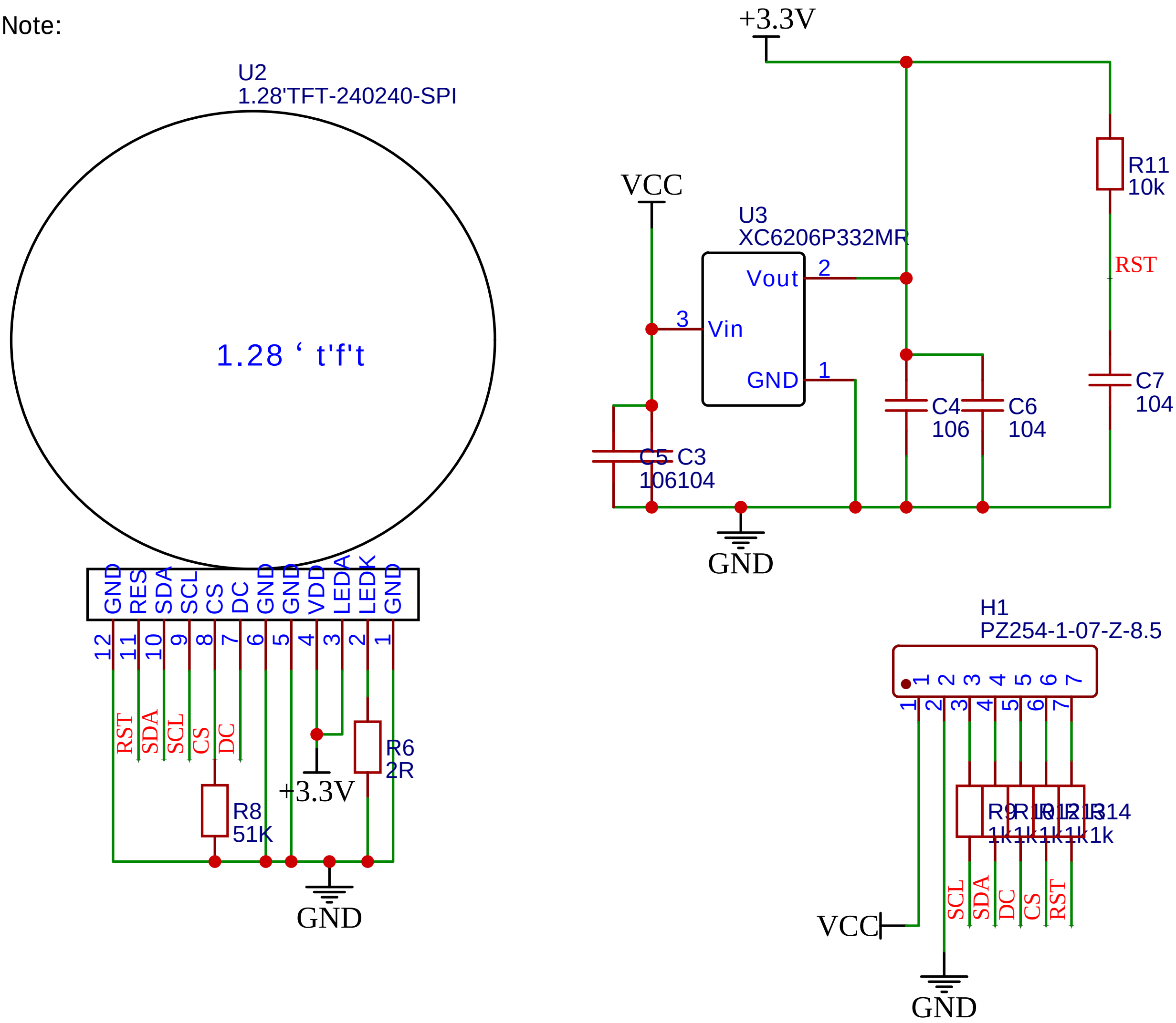
MODEL NUMBER :	SCALE:	UNITS:
GMT128-02_LCM	Unspecified TOL:± 0.2	MM
	DO NOT SCALE THIS DRAWING.	

3. Pin Definition

FPC Connector is used for the module electronics interface.

NO.	Symbol	Description
1	VCC	Power
2	GND	Power Ground
3	SCL	This pin is used to be serial interface clock
4	SDA	SPI interface input/output pin. the data is latched on the rising edge of the SCL signal.
5	DC	Display data/command selection pin in 4-line serial interface.
6	CS	Chip selection pin;Low enable, high disable.
7	RST	This signal will reset the device and it must be applied to properly initialize the chip. Signal is active low.

Note:



4. Electrical Characteristics

4.1 Absolute Maximum Ratings

Parameter	Symbol	Min	MAX	Unit	Notes
Supply Voltage (I/O)	VDD	-0.3	4.6	V	
Analog Supply Voltage	VDDIO	-0.3	4.6	V	
Logic Input Voltage	VIN	-0.3	VDDIO+0.3	V	
Operation Temperature	Top	-20	70	℃	
Storage Temperature	Tst	-30	80	℃	

4.2 Operating Conditions

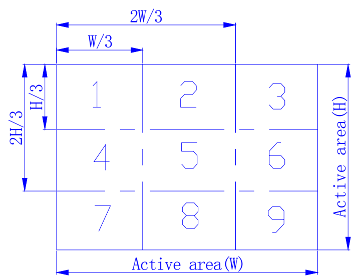
Parameter	Symbol	Min	TYP	MAX	Unit	Notes
System Voltage	VDD	2.5	2.8	3.3	V	
Interface Operation Voltage	VDDIO	1.65	1.8	3.3	V	
Gate Driver High Voltage	VGH	12	-	13	V	
Gate *Driver Low* Voltage	VGL	-11	-	-8	V	
Operating Current for V_{DD}	I_{DD}	-	8.5	10.5	mA	
Sleep_In Mode VDD	I_{dd}	-	15	30	uA	
Sleep_In Mode VDDIO	I_{ddio}	-	5	10	uA	

4.3 Backlight Unit

Parameter	Symbol	Min	TYP	MAX	Unit	Notes
Voltage for LED backlight	VLED	2.9	3.0	3.1	V	
Current for LED backlight	ILED	-	40	60	mA	2 LED Parallel
Power Consumption	Pbl	-	120	186	mW	1
Brightness	L_{br}	350	400	-	cd/m^2	2
LED Life time	-	20000	-	-	hr	3

Note:

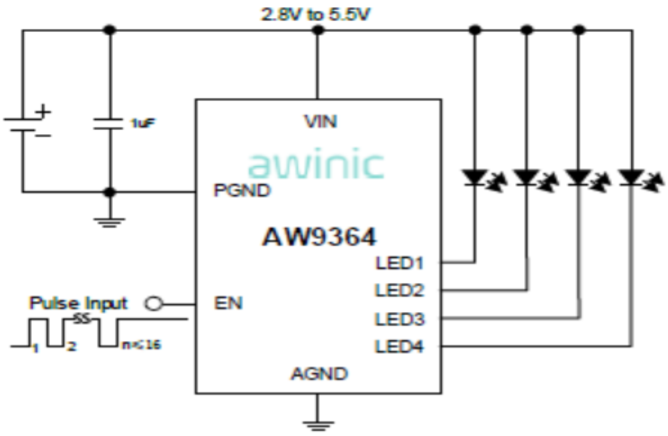
1. Where ILED =40mA , VLED=3.0V , Pbl= ILED x VLED
2. Uniform measure condition:
   a:Measure 9 point,Measure location is show below:
   b:Uniform=(Min brightness/Max.brightness)x100%
   c:Best Contrast.



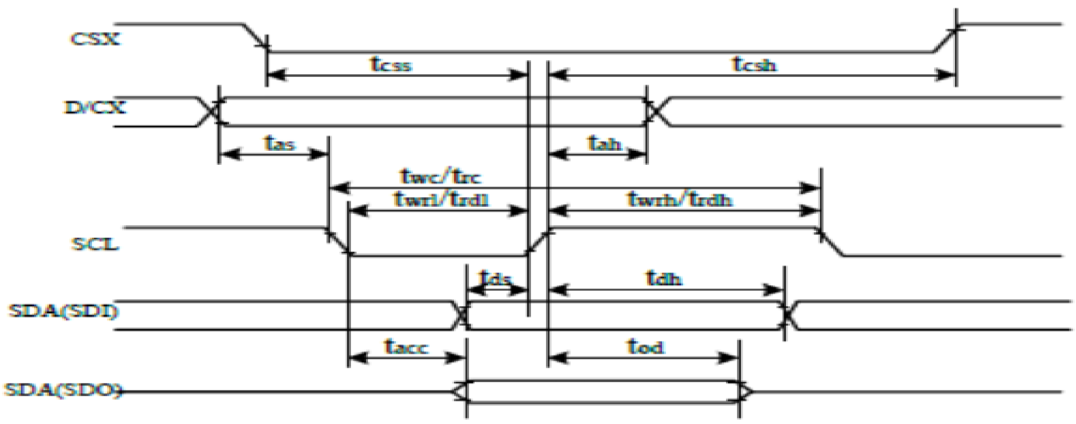
3. The environmental conducted under ambient air flow ,at Ta=25±2^{o}C,60%RH±5%

4.4 Backlight Recommended Circuit

Motherboard driver backlight is need constant current circuit , if threated voltage screen after light brightness difference . Current and power consumption of the machine are inconsistent , so recommend a backlight driving circuit is best rated current . It is recommended to use IC (AW9364) . The reference circuit is as follows:



4.5 AC Timing Characteristic of The LCD



Signal	Symbol	Parameter	min	max	Unit	Description
CSX	tcss	Chip select time (Write)	20	-	ns	
	tcsh	Chip select hold time (Read)	40	-	ns	
SCL	twc	Serial Clock Cycle (Write)	10	-	ns	
	twrh	SCL "H" Pulse Width (Write)	5	-	ns	
	twrl	SCL "L" Pulse Width (Write)	5	-	ns	
	trc	Serial Clock Cycle (Read)	150	-	ns	
	trdh	SCL "H" Pulse Width (Read)	60	-	ns	
	trdl	SCL "L" Pulse Width (Read)	60	-	ns	
D/CX	tas	D/CX setup time	10	-	ns	
	tah	D/CX hold time (Write/Read)	10	-	ns	
SDA/SDI (Input)	tds	Data setup time (Write)	5	-	ns	
	tdh	Data hold time (Write)	5	-	ns	
SDA/SD0 (Output)	tacc	Access time (Read)	10	-	ns	